

SANJANA UMESH SAWANT

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SUMMARY

Data science graduate building practical data and AI products across analytics, KPI reporting, ETL, applied modeling, ranking systems, and product-facing experimentation. Strongest at turning messy data into usable insight and translating technical work into clear business value. Targeting data analyst, data scientist, AI, analytics engineering, and fintech-oriented roles.

EDUCATION

Rutgers University | M.S. in Data Science, High Honor **May 2026**
GPA: 3.6/4.0 | Coursework: Machine Learning, Regression and Time Series, DBMS, Deep Learning, LLMs, Agentic AI
SRM Institute of Science and Technology | B.Tech. in Electronics and Computer Engineering **Jun 2024**
GPA: 9.01/10

EXPERIENCE

Helius Health AI | Data Science Intern | Remote **Spring 2026**

- Enabled production loading of validated healthcare workforce data by building and testing local ETL/ELT workflows that staged raw survey files into Supabase/PostgreSQL and were reviewed by the CTO before handoff.
- Gave sponsors visibility into onboarding, assessment completion, engagement, retention, conversion, drop-off, and specialty matching by querying Supabase/PostgreSQL through API access and turning outputs into KPI analyses.
- Made one-run monitoring of user and workforce activity possible by building an end-to-end KPI reporting workflow with graphical outputs for sponsor-facing engagement and onboarding metrics.
- Advanced a nursing specialty-matching decision-support model toward review by developing an initial predictive workflow that estimates which department may best fit each nursing student.

NeuralSeek | AI Agent Developer Intern | Remote **Spring 2026**

- Developed an AI-powered restaurant recommendation workflow using retrieval-based reasoning and recommendation-ranking logic.
- Analyzed user feedback and recommendation behavior to identify factors influencing trust, relevance, and decision quality in AI-assisted discovery.
- Validated the product with users and increased willingness to trust AI-generated recommendations from 17% to 78%, then extended the work into InstaDine.

Superpack Packaging Machines Pvt. Ltd. | PLC Programming Engineering Intern **Oct 2023 - Jul 2024**
Hyderabad, India

- Improved industrial filling-machine control logic for fluids with different viscosity levels through testing and engineering iteration, increasing machine efficiency by 4% to 8%.

SELECTED DATA SCIENCE PROJECTS

InstaDine | Full-Stack AI Restaurant Recommendation Platform **Spring 2026**

- Built a full-stack AI product using Flask, PostgreSQL, Docker, Google Places API, and Vertex AI/Gemini to deliver personalized, live-data-grounded recommendations.
- Designed conversational retrieval, ranking, filtering, session, and persistence workflows combining user preferences, travel constraints, ratings, and availability.

Loan Default Prediction | Credit Risk Modeling **Summer 2026**

- Built an end-to-end credit-risk modeling project to predict loan default from 307K+ borrower records using Python, pandas, scikit-learn, and XGBoost.
- Compared logistic regression and XGBoost with ROC-AUC, PR-AUC, precision/recall, calibration, and threshold optimization to connect model performance to lending tradeoffs.

Bankruptcy Prediction Pipeline and Explainer App **Summer 2026**

- Built reusable preprocessing and modeling pipelines for scaling, encoding, missing-value imputation, interpretable classification, artifact persistence, and rare-event evaluation.
- Framed the workflow through a business-friendly frontend explaining why naive accuracy is misleading for imbalanced financial-risk prediction.

Quantifying the Environmental Cost of AI **Fall 2025**

- Compared transformer fine-tuning tradeoffs across DistilBERT, BERT-family models, and GPT-style baselines using F1 performance and carbon-emissions cost.
- Analyzed full fine-tuning, LoRA, and efficient training setups, finding roughly 78% lower emissions with limited performance loss.

Effect of Technology Use on Mental Health **Spring 2025**

- Processed NTIA and NSDUH survey data across multiple years, standardized inconsistent historical formats, and built R/Databricks transformation workflows.
- Identified demographic trends while clearly separating correlation from causation in the final analysis.

TECHNICAL SKILLS

Python, R, SQL, pandas, scikit-learn, XGBoost, PyTorch, TensorFlow, PySpark, Flask, FastAPI, Streamlit, PostgreSQL, Supabase, SQLite, Databricks, BigQuery, Power BI, Docker, Git, REST APIs, OpenAI API, LLM evaluation, prompt engineering, LoRA fine-tuning, data pipelines, ETL, KPI analytics, experimentation, model evaluation, credit-risk modeling